

Year 9 Science – Booklet 9: Biology

Human Systems — ANSWERS

Quick Test (p.3)

1. It is the organ of photosynthesis – it makes food for the plant.
2. Xylem (tubes).
3. The pancreas.

Section A – Multiple Choice (5 marks)

1. b) xylem
2. d) the leaf
3. d) holes on the underside of the leaf
4. c) produces hormones
5. c) kidney

Section B (10 marks)

1. Label the flowering plant diagram and state each part's function (8 marks)

A – **Flower**: contains the male and female sex organs and makes seeds (and is usually brightly coloured to attract insects for pollination).

B – **Stem**: holds the plant upright and contains the xylem and phloem tubes that transport substances around the plant.

C – **Root**: anchors the plant in the soil, and takes up water and minerals from it.

D – **Leaf**: the organ of photosynthesis; it makes food for the plant.

(Check the letters against your own diagram – positions can vary between printings.)

2. Names of the tubes (2 marks)

Xylem tubes carry water and nutrients. **Phloem** tubes carry glucose.

Section C (7 marks)

1. Kidney and bladder diagram

- a). The excretory system.
- b). The bladder.

2. Sense organs and other body systems

- a). The nervous system.
- b). Via nerves, which carry (electrical) signals from the sense organs to the brain and spinal cord – together forming the central nervous system.
- c). Example (3 marks): the digestive system – organs include the mouth, stomach, small intestine and large intestine (plus the liver and pancreas); its function is to break food down into small, soluble molecules that can be absorbed into the blood for energy and growth, and to remove indigestible waste. (Other valid systems: respiratory – lungs, trachea; gas exchange. Circulatory – heart, blood vessels; transports substances around the body. Skeletal – bones; support, protection, movement.)